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Optimal mass minimization design of a two-stage coaxial helical speed reducer with Genetic Algorithms

Ovidiu Buiga *, Lucian Tudose [1]

Technical University of Cluj-Napoca, Faculty of Machine Building, Department of Mechanical Engineering Systems, Bvd. Muncii 103-105, 400641 Cluj-Napoca, Romania

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ABSTRACT

The full description of a two-stage speed reducer generally requires a large number of design variables (typically, well over ten), resulting a very large and heavily constrained design space. This paper presents the specific case of the complete automated optimal design with Genetic Algorithms of a two-stage helical coaxial speed reducer. The objective function (i.e. the mass of the entire speed reducer) was described by a set of 17 mixed design variables (i.e. integer, discrete and real) and also was subjected to 76 highly non-linear constraints. It can be observed that the proposed Genetic Algorithm offers better design solutions as compared with the results obtained by using the traditional design method (i.e. a commonly trial and cut error).

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1. Introduction

The design problem of a mechanical power transmission such as a two-stage helical coaxial speed reducer is a complex process involving a number of challenges, particularly when considering the multiple interdependencies between the design variables defining its subsystems (i.e. the gearings, the shafts-subassembly, and the housing-case and cover). In other words, an optimal reducer is generally not an assembly of components optimized in isolation, a fact overlooked by many conventional design heuristics [1]. The optimization of the complete speed reducer leads to a complicated objective function, with a large number of design variables of mixed nature (i.e. integer values-for gears number of teeth, discrete values-for normal tooth addendum coefficients and real values-for gears width) and highly non-linear constraints. Moreover to design means to make decisions, which unfortunately are always compromises. A typical example might be that selecting a smaller than optimal gear diameter (and a correspondingly greater contact width) could yield a somewhat heavier gearing, but a more compact layout and therefore a much lighter housing; it is worth mentioning though that in reality the impact on the overall objective tends to be much less direct and therefore much more obscure than in this example [1]. It is clearly that it is almost impossible to tell what the first compromise should have been, let alone what any subsequent choices should have been made with the overall goal in mind. If we take into account all the above facts, it is obviously that the traditional design (for such a complex problem) is a very difficult process, so the computer aided design of the gears is needed. Anyway, a more pronounced increase demands for compact, efficiency, and reliable gears have forced the designer to use the optimal design methodology [2]. In the last decades many researchers have paid attention on this problem of gear optimization. Madhusudan and Vijayasimha in [3] presented a computer program in order to design a required type of gear under a specified set of working conditions. In [4] a new computer-aided method for automated gearbox design is described. Huang et al. developed an interactive physical programming in order to optimize a three-stage spur gear reduction unit [5]. Abersek et al. in [6] described an expert system to design and manufacture a gearbox. Li et al. [7] reported a study for minimizing the center distance of a helical gear using American Gear Manufacturers Association (AGMA) procedures. Yokota et al. in [8] solved an optimal weight design problem of a gear with an improved Genetic Algorithm (GA). Deb and Sachin, in [9] used a non-dominated sorting Genetic Algorithm (NSGA-II) in order to solve a multi-objective optimization of a multi-speed gearbox. Thompson et al. [10] presented a generalized optimal design of two-stage and three-stage spur gear reduction units in a formulation with multiple objectives. Ray and Saini illustrated in [11] the benefits of the particle swarm searches in resolving different engineering designs. In [2] Savsani et al. presented two advanced optimization algorithms known as particle

* Corresponding author. Tel.: +40 264401753 (Office), mobile: +40 746105871.
E-mail addresses: Ovidiu.Buiga@omt.utcluj.ro (O. Buiga), lucian.Tudose@omt.utcluj.ro (L. Tudose).
[1] Tel.: +40 264401665 (Office), mobile: +40 745640604.

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swarm optimization (PSO) and simulated annealing (SA) in order to minimize the weight of a spur gear train. The results of the proposed algorithms were compared with the results obtained by Yokota et al. [8]. Gologlu and Zeyveli in [12] applied GA to minimize the volume of a two-stage helical gear train. A design methodology for two-stage speed gearboxes is presented in [13]. These studies referenced above have been instrumented in order to highlight two important aspects regarding the topic of this work.

Firstly, it is obvious that all the authors cited above applied the optimization techniques for individual components (only gears or shafts) or intermediate assemblies rather than to the entire speed reducer (considering the multiple connections between its subsystems, i.e. the gearings, the shafts-subassembly, and the housing). In the same vein and in order to find out which are the dimensional tendencies of the speed reducer's components and how the functional and structural interdependencies affect them we decomposed the optimal design problem into a set of *two tractable subproblems*. So we dealt separately with the *gearings* [14] and *the shafts* [15]. In this way we could take the correct and necessary decisions for the complex and difficult *optimal design of the complete speed reducer* [1]. In [1] we minimized the mass of the entire speed reducer using a two-phase evolutionary algorithm, based on the paradigm of '*punctuated equilibra*', i.e. a phenomenon associated by some biologists with the so-called *Cambrian explosion*, a time of rapidly increasing genetic diversity.

Secondly, these studies emphasize the importance of using modern global optimization techniques in the mechanical power transmission design (as opposite to conventional, trail and error type methods), even when considering certain components (i.e. gears, shafts etc.).

In order to ***build a generic transmission system design tool based on the evolutionary optimization concepts*** [1] in this paper we successfully managed to optimal design a *new*, *complex* and *interesting layout* (see for example Fig. 1) of a two-stage speed reducer (i.e. a *coaxial one*-which means the input and output shafts are *collinear* – Fig. 2). This new layout introduces a number of challenges (considering the center distance for both stages has the same value) as opposite to the example presented in Tudose et al. [1]. Moreover, as opposite to previous reminded optimal design, in here we considered new materials for all four gears (i.e. case hardened alloy steel as it can be observed in Section 5) and a new set of input data.

We shall now introduce the traditional speed reducer design method (currently used for designing a multi-stage power transmission – Section 2), after which we describe the general principle of the proposed Genetic Algorithm (Section 3), followed by a detailed discussion regarding the statement of the optimal design problem (the objective function, the design variables and the constraints – Section 4). The fifth Section contains an effective example and a detailed presentation and comparison of the numerical results for optimal and traditional design (i.e. a commonly trial and cut error procedure) solutions. Eventually, we conclude this discussion with some reflections and suggestions regarding the possible extensions of the present study.

2. Traditional speed reducer design overview

The traditional speed reducer design process depends on the designer's intuition, experience, and skills [16]. The flowchart of a traditional speed reducer design is presented in Fig. 1. As one can notice, there are 3 distinct, though interdependent sections, necessaries to design the entire speed reducer. In the following a description of each section is presented. The first section consists in **gearing design**. The main difficulty that arises in traditional gear set design lies in the fact that it is necessary to know all the dimensions of the gears, as well as the tooth form and size, before the loads and stresses may be accurately determined. This makes it necessary to *estimate* the size of the gears (i.e. steps -1*a*- through -1*i*- from Fig. 1), using simplified methods, and then to *check* the estimate, using the various design factors (K_v, $K_{H\beta}$, $K_{H\alpha}$, Z_ε, Z_H, Z_β, $K_{F\beta}$, $K_{F\alpha}$, Y_{Fa}, Y_{Sa} and Y_β; German Industrial Standard (DIN) 3990 [17]) in conjunction with more exact equations (i.e. steps -1*j*- through -1*r*- from Fig. 1). The estimated size and tooth form are then altered in accordance with the information obtained from the exact equations [18]. Let us now present the steps (pointed out in the above rows) used in gear design [17,19,20]. It is well

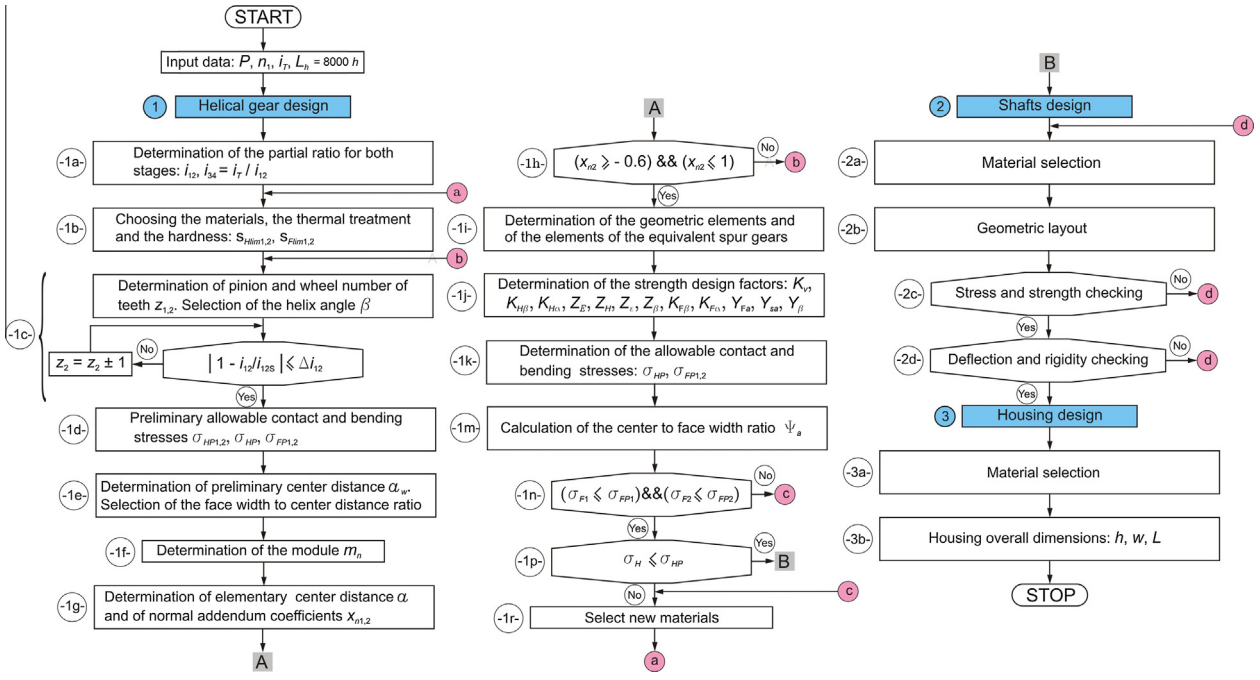


Fig. 1. The flowchart of a traditional speed-reducer design.

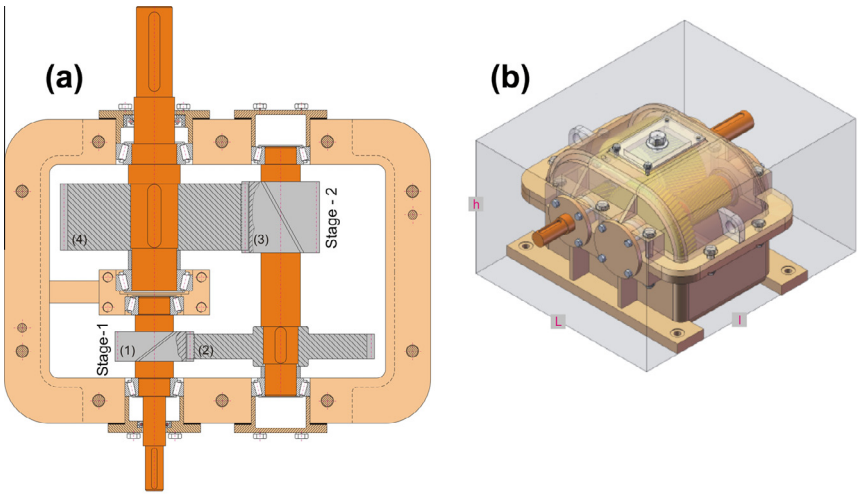


Fig. 2. (a) Section of the two-stage coaxial helical speed reducer; and (b) 3D view of the speed reducer.

known that in a speed reducer unit initial input speed n_1 and output speed n_3 determine the total gear ratio i_T [12]. Thus, the process of speed reducer design begins with determination of the partial gear ratios for both stages (-1*a*-). Then, the gearing of the first stage is computed starting by choosing the materials, the hardness, and the thermal treatment for pinion and wheel, respectively (-1*b*-). Next, the number of teeth on pinion and gear are determined. The actual value of the gear ratio should not differ from the nominal ones by more than 4% (for double reduction units according to [19]) (-1*c*-). National Romanian Standard (i.e. STAS 6012) offers a more restrictive value for the actual gear ratio (which should be in the range of ±2.5%). This value was used in this paper in order to check the relative error of the actual gearing ratio (i.e. constraints C1 and C2 from Section 4.3). The estimated allowable contact ($\sigma_{HP1,2}$, σ_{HP}) and bending stresses ($\sigma_{FP1,2}$) for the gears are then computed (-1d-). In the following step the preliminary center distance a_w is determined and the gear width to center distance ratio ψ_a is selected (i.e. 0.25,. . .,0.4, depending on the position of gears – [19]) (-1*e*-). Then, the normal module m_n (-1*f*-), the elementary center distance a and the normal teeth addendum coefficients $x_{n1,2}$ (-1g-) are determined. Then, the elements of the helical gears and of the equivalent spur gears are calculated (-1*i*-). All the strength design factors are exactly computed (-1*j*-). The allowable contact and bending stresses (-1*k*-) are calculated as well as the face width to center distance ratio (-1*m*-). The gearing is checked on bending (-1*n*-) and contact (-1*p*-) stresses. If the calculated values are lower than the allowable ones (-1*m*-), then the former parameters of gearing are taken as final. Otherwise, recalculation is required. From this brief description it can be noticed that gear kinematics, geometry and strength related aspects conduct to a complicated and time consuming gear design. The following section describes the **shaft design**. Since fatigue shaft design depends so heavily on the material choice [20], first a reasonable material selection should be performed (-2*a*-). The general layout of the shaft, including the axial location of gears and tapered roller bearings must be specified (-2*b*-) in order to perform a free-body force analysis and to obtain shear force and bending moment diagrams. Decisions on how to transmit the torque from the gears to shaft (selection of various types of keys) need to be made as well as how to hold gears and tapered roller bearings in place. Next steps (-2*c*-) consist in shaft design for stress (fatigue and static-at critical location) and for deflection (-2*d*-). The last section of the speed reducer traditional design concerns the sizing of its **housing**. This starts once that the virtual design of each stage is generated. Next the material is chosen (-3*a*-) and all the specific elements of the housing (i.e. flanges, ribs, bossage, side covers, etc.) are determined (-3*b*-).

A traditional design for a multi-stage speed reducer (as it can be seen from the above brief description) involves computations based on contact stress, bending strength, interference etc., leading to a complex and time-consuming process. Moreover the design so obtained may not be the optimum [2]. To overcome these drawbacks we proposed a GA. We consider that GAs represent the direction of choice at the present in solving such complex design problem. In the following section a short description of the GA will be presented.

3. The Genetic Algorithm

The Genetic Algorithm is perhaps the most well-known of all evolution-based search algorithms [21], suited for solving complex optimization problems [22]. The basic concepts of GAs were developed by Holland [23]. GAs maintain a population of solutions, then allow the fitter individuals to reproduce, and let the less fit individuals die off [22]. Each individual consists of a *genotype* (i.e. the search space of coded solutions) and a corresponding *phenotype* (i.e. the solution space). Phenotypes usually are collections of parameters. Genotypes consist of coded versions of these parameters (referred to as a *gene*). A collection of genes in one genotype is often held internally as a string, and is known as a *chromosome* [22] (the chromosome of our two-stage coaxial helical speed reducer is composed from 17 design variables-as it can be seen in Section 4).

This algorithm works as follow: (*a*) the genotype (i.e. the search space of coded solutions-which in our case is vast: 1.5×10^{23} – representing the possible optimal speed reducer designs) of every individual in the population is randomly initialized. (b) The phenotype (i.e. collections of parameters which in this paper might define the number of teeth on pinions, the standardized center distance, the radial input/output shafts sealing, the tapered roller bearings which support the input/output shafts etc.) of every individual from the initial population is evaluated using the fitness function (i.e. the mass of the complete speed reducer). Next, (c) two parents are randomly selected (using the roulette wheel method) for reproduction based on the fitness values of the individuals. Offsprings are created (*d*, *e*) by applying the genetic operators: crossover (merges information from two parents into one or two offsprings) and mutation (acts on a single offspring and works by

applying some variation to one or more genes in the offspring's chromosome). The new generated individuals are evaluated (f) using the fitness measure. After the evaluation, the offspring replaces some/all of the individuals in the current population (g). This full process of evaluation and reproduction continues until either a satisfactory solution emerges or the GA has run for a specified number of generations.

4. Statement of the optimal design problem

4.1. The 'genotype' of the two-stage coaxial helical speed reducer

The first step in the formulation of the optimal design problem is to identify the set of *genes* (design variables) that uniquely describe the two-stage coaxial helical speed reducer (Fig. 2). This mechanical power transmission is highly standardized. The 17 design variables that unequivocally define the reducer reflect this [1]. As one can observe in Table 1, from the set of 17 genes, 9 are standardized.

4.2. The objective function

Mass is an important machine parameter. In general engineering a reduced weight of machines means metal economy and cheaper production. Taking these aspects into account, the minimization of the two-stage coaxial helical speed reducer mass is the objective function of this study. The mass is defined as shown in the following equation.

$$\begin{aligned}\text{Obj}: \quad & F(x) \\ &= \sum_{i=1}^{5}(V_{11i}+V_{12i})\rho_1 + \sum_{j=1}^{2}\left(V_{21j}+V_{22j}+V_{23j}+V_{24j}\right)\rho_2 \\ &\quad + \sum_{k=1}^{6}(V_{31k}+V_{32k}+V_{33k})\rho_2 \\ &\rightarrow \min \end{aligned} \tag{1}$$

where ρ_1 is the density of cast iron (i.e. 7.2×10^{-6} mm^3/kg) and ρ_2 is the density of steel (i.e. 7.85×10^{-6} mm^3/kg). The components of the volumes corresponding to the case and cover (V_{11i} and V_{12i}), gearings (V_{21j}, V_{22j}, V_{23j} and V_{24j}) and shafts (V_{31k}, V_{32k} and V_{33k}) are shown in Fig. 3.

4.3. Constraints

In this section, the constraints for the two-stage coaxial helical speed reducer optimal design are described. There are a total of 76 constraints (involving strength, geometrical and structural considerations) typically encountered in practical design of a two-stage speed reducer. These constraints divide the design space into two domains, the *feasible domain* (where all the constraints are satisfied), and the *infeasible domain* (where at least one of the constraints is violated). All these constraints are of inequality type, defined as $g_i = a_i/b_i - 1 \leqslant 0$, where a constraint is of the form $a_i \leqslant b_i$, with a_i, $b_i > 0$. It is obviously that the values of all these constraints have to be negative or at last zero (case in which the solution is feasible). For the sake of conciseness, we shall not dwell on the details regarding their calculation. It should be mentioned here that the housing of the speed reducer (containing its specific elements, i.e. flanges, ribs, bossage, etc.) was also designed automatically once the virtual design of the gearings was generated. This it is absolutely necessary in order to enable the calculation of some constraints (e.g. the operating temperature, lubrication condition, and some geometrical constraints). These 76 constraints were divided into 5 groups which should be with reference to the sketch presented in Fig. 2.

Constraint group 1 – *regarding the* ***gearings*** *of the speed reducer.*

The set of **37 constraints** from this group are based on the gearings calculations (which could be founded in the German Industrial Standard (DIN) 3990 [17]). These constraints are described below. **C1**, **2** – *The relative error of the actual gearing ratio on the both stages must be within the range of* [−2.5%,. . .,+2.5%]. **C3**, **4** – *The Hertzian contact pressure on the teeth of gears on both stages must not exceed the allowable Hertzian contact pressure.* **C5–8** *The bending stress on the teeth of gears on both stages must not exceed a specified value.* **C9–12** *The teeth of all* 4 *gears of the two-stage coaxial helical speed reducer must not be undercut.* **C13–16** *The top land on the teeth on gears* 1 *through* 4 *must not vanish.* **C17**, **18** *The contact ratio on first and second stage must be greater than a specified value* (i.e. the minimum front contact ratio $\varepsilon_{\alpha min} = 1$). **C19**, **20** – *The addendum coefficient of the wheel* (2) *and* (4) *should be in the range of* [−0.6,. . .,1]. **C21**, **C22** – *The numbers of teeth on both stages must be relative primes.* **C23–34** *A set of measurability constraints (i.e. span measurement) of the gears on both stages.* **C35–37** *Constraints related to gears manufacturability.*

Constraint group 2 – *about the* **shafts sub-assembly**.

This group is composed by a set of **12 constraints** regarding the interconnections between the shafts sub-assembly components (i.e. end-shafts, radial seals and tapered roller bearings) and **17 more constraints** about shafts ***stress*** and ***strength*** (i.e. static strength: C50–52; and fatigue strength: C53–55); and ***deflection*** (i.e. bending deflection: C56–60; torsional deflection: C64–66; and deflection at the supporting points: C61–63). The components of shafts sub-assembly are design variables (as it can be observed in Table 1). They are coded as integer variables and offer important details about the *geometrical* and the *mounting dimensions* of the

Table 1
The 17 genes describing the two-stage coaxial helical speed reducer.

Symbol	Range	Description
i_{12S}	{1.12,. . ., 40}	Gear ratio of the 1st stage. Standardized, discrete real values
z_1, z_3	{14,. . ., 21}	Number of teeth of 1st and 2nd stage pinions, respectively. Integer values
a_w (mm)	{56,. . ., 315}	Center distance. Standardized, discrete, real values
x_{n1}, x_{n3}	{−0.6,. . .,1}	Normal tooth addendum coefficients of the 1st and 2nd stage pinions, respectively. Discrete, real values
β_{1st}, β_{2st} ($^{\circ}$)	[4, 19.75]	Helix angle measured at the pitch diameters of the 1st and 2nd stage, respectively. Discrete real values
ψ_{a1st}, ψ_{a2st}	[0.2,. . .,0.8]	Gear width to center distance ratio coefficient of the 1st and 2nd stage, respectively. Real values
i_1	{0,. . .,63}	Index of standardized end for the input shaft. Integer values
i_2	{0,. . .,127}	Catalogue index of radial shaft seal for the input shaft. Integer values
i_3	{0,. . .,127}	Catalogue index of tapered roller bearings for the input shaft. Integer values
i_4	{0,. . .,127}	Catalogue index of tapered roller bearings for the intermediary shaft. Integer values
i_5	{0,. . .,63}	Index of standardized end for the output shaft. Integer values
i_6	{0,. . .,127}	Catalogue index of radial shaft seal for the output shaft. Integer values
i_7	{0,. . .,127}	Catalogue index of tapered roller bearings for the output shaft. Integer values

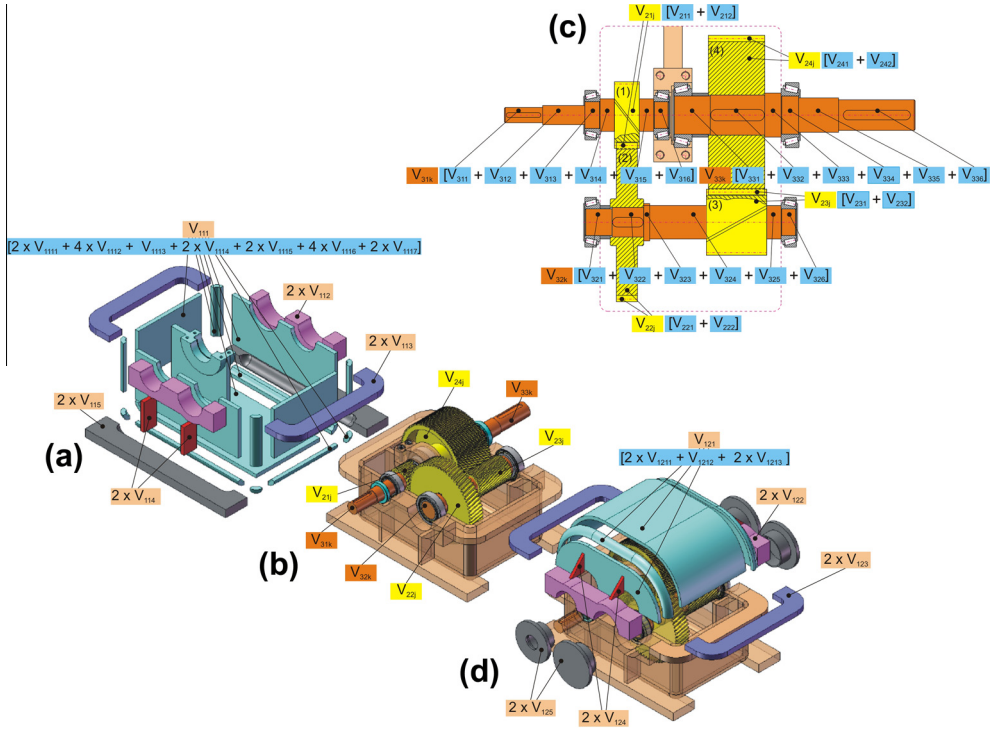


Fig. 3. (a) Exploded view of case of the reducer; (b and c) sub-assembly of the reducer's shafts; and (d) exploded view of cover of the speed reducer.

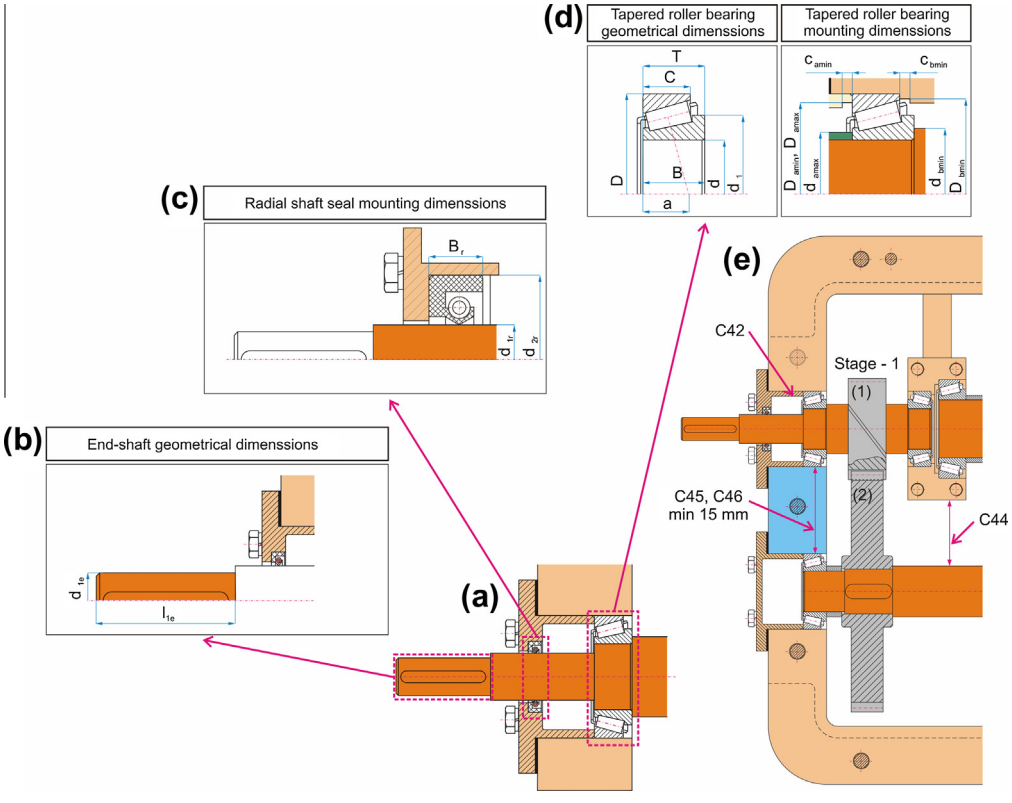


Fig. 4. (a) Input end shaft sub-assembly; (b) input end shaft geometrical dimensions; (c) radial shaft seal mounting dimensions; (d) tapered roller bearing geometrical/mounting dimensions; and (e) constraints C42, C44, C45 and C46 illustrations.

end-shafts (Fig. 4b), radial-seal (Fig. 4c) and bearings (Fig. 4d). In Fig. 4e are also pointed out the constraints C42–C43, C44 and C44–C45. In the following rows are described the constraints that belong on this group. **C38**, **39** – *The input and output shaft ends must have sufficient diameter step to allow the mounting of a belt wheel.* **C40**, **41** – *The mounting diameter of the radial seal must be less than*

Table 2
The values of the genes obtained after optimization.

Symbol	i_{12S}	z_1	z_3	a_w (mm)	x_{n1}	x_{n3}	β_{1st} (°)	β_{2st} (°)	ψ_{a1st}	ψ_{a2st}	i_1	i_2	i_3	i_4	i_5	i_6	i_7
Value	3.15	20	21	90	1	−0.393	18	17.25	0.294	0.781	12	26	5	6	32	78	28

Table 3
Gearing comparison for both design solutions.

Denotation	Traditional design	Optimal design
1st Stage		
Number of teeth on pinion, z_1	17	20
Number of teeth on wheel, z_2	60	63
Actual gear ratio, u_{12}	3.524	3.15
Normal module, m_{n1st} (mm)	2.5	2
Pinion width, b_1 (mm)	36	30
Wheel width, b_2 (mm)	32	26
Pinion root diameter, d_{f1} (mm)	38.007	40.458
Wheel root diameter, d_{f2} (mm)	148.539	128.872
Pinion working diameter, d_{w1} (mm)	44.155	43.373
Wheel working diameter, d_{w2} (mm)	155.844	136.626
Pinion outside diameter, d_{a1} (mm)	49.46	49.527
Wheel outside diameter, d_{a2} (mm)	159.992	137.941
2nd Stage		
Number of teeth on pinion, z_3	21	21
Number of teeth on wheel, z_4	43	47
Actual gear ratio, u_{34}	2.047	2.238
Normal module, m_{n2st} (mm)	3	2.5
Pinion width, b_3 (mm)	57	74
Wheel width, b_4 (mm)	53	70
Pinion root diameter, d_{f3} (mm)	58.725	46.004
Wheel root diameter, d_{f4} (mm)	124.811	120.069
Pinion working diameter, d_{w3} (mm)	65.625	55.588
Wheel working diameter, d_{w4} (mm)	134.375	124.411
Pinion outside diameter, d_{a3} (mm)	72.788	57.93
Wheel outside diameter, d_{a4} (mm)	138.874	131.995

the inside diameter of the tapered roller bearings on the input and output shafts. **C42**, **43** – *Geometrical constraint related to the space required by the outside ring of the tapered roller bearings on the input and output shafts.* **C44** *The non-interference condition between the tapered roller bearings housing of the intermediary and output shafts.* **C45**, **46** – *The minimum distance between adjacent tapered roller bearings must be greater than* 15 *mm.* **C47–49** *The basic service life* (*according to* ISO 281:2007) *of the tapered roller bearings must exceed a specified value* i.e. *the specification life of* 15,000 operating hours. **C50–52** *Equivalent von Misses maximum stresses experienced by all the three shafts should be less then the allowable bending stresses.* **C53–55** *The fatigue life safety factors on the input, output and intermediary shaft must not fall below a specified value* (the fatigue strength was computed according Soderberg [20], and the specific value is 1.5). **C56–60** *The bending deflection on the three shafts in the critical locations must be below certain threshold values* (i.e. $0.001 \cdot m_n$, where m_n is the normal module) *to enable the correct functioning of the gearings and the tapered roller bearings.* **C61–63** *The deflection at the supporting point angle of all three shafts should be less then an allowable value* (i.e. 1.2×10^{-3} rad, [20]). **C64–66** *The torsional deflection of the three shafts must be below a threshold value* i.e. 0.436×10^{-5} rad/mm.

Constraint group 3 – *regarding the* **key joints**.

C67–70 *The shearing and crushing stresses must not exceed a specified value (i.e. allowable shearing stress* 70 MPa; *and allowable bearing stress* 80 MPa, *corresponding to the material of the keys i.e. steel grades E*355 (*EN*), [19]) *on the keys and keyways of the input and output end shafts.* **C71–74** *The shearing and crushing stresses must not exceed a specified value on the key and keyway for mounting the wheels* (2) *and* (4).

Constraint group 4 – *about the* **lubrication of the gearings**.

C75 – *The margin between the minimum and maximum allowable lubricant levels should be no less than* 10 mm.

Constraint group 5 – *regarding the* **thermal design** *of the speed reducer.*

C76 – *The operating temperature of the two-stage coaxial helical gear unit must not exceed a specified value (i.e. the permissible operating temperature* 65 °C).

5. A two-stage coaxial helical speed reducer mass minimization example

This is probably a good time to consider the practical example of a 3 kW two-stage coaxial helical speed reducer optimal design. The objective is to obtain a minimum mass speed reducer for a service life of 8000 h (which satisfies the list of the constraints presented in Section 4) given an input speed of 750 rpm and a total transmission ratio of 7.1. The gears of the speed reducer should

Table 4
Comparison of shafts for both design solutions.

Denotation	Traditional design	Optimal design
Input shaft		
End shaft diameter, d_{es1} (mm)	20	18
Radial sealing	CR 24 × 37 × 7 HMS5 RG	CR 21 × 35 × 7 HMS5 RG
Radial sealing inner diameter, d_{1rs1} (mm)	24	21
Tapered roller bearing	32,005 X/Q	32,005 X/Q
Tapered roller bearing bore diameter, d_{trb1} (mm)	25	25
Input shaft total length, l_{s1} (mm)	162	147
Intermediate shaft		
Tapered roller bearing	30,205 J2/Q	30,205 J2/Q
Tapered roller bearing bore diameter, d_{trb2} (mm)	25	25
Intermediate shaft total length, l_{s2} (mm)	199.75	203.5
Output shaft		
End shaft diameter, d_{es3} (mm)	30	32
Radial sealing symbol	CR 35 × 55 × 10 HMS5 RG	CR 36 × 58 × 10 HMS5V
Radial sealing interior diameter, d_{1rs3} (mm)	35	36
Tapered roller bearing symbol	32,207 J2/Q	32,008 X/Q
Tapered roller bearing bore diameter, d_{trb3} (mm)	35	40
Output shaft total length, l_{s3} (mm)	241.25	226

Table 5
Comparison of some important characteristics of the speed reducer for both design solutions.

Denotation	Traditional design	Optimal design
Maximum Hertzian contact pressure in the teeth of gears on 1st stage, σ_{H1st} (MPa)	671.362	718.589
Allowable Hertzian contact pressure in the teeth of gears on 1st stage, σ_{HP1st} (MPa)	713.042	725.272
Maximum Hertzian contact pressure in the teeth of gears on 2nd stage, σ_{H2st} (MPa)	732.254	712.682
Allowable Hertzian contact pressure in the teeth of gears on 2nd stage, σ_{HP2st} (MPa)	720.882	714.269
Bending stress on teeth of the pinion, σ_{F1} (MPa)	122.362	120.866
Allowable bending stress on teeth of the pinion, σ_{FP1} (MPa)	571.574	576.312
Bending stress on teeth of the wheel, σ_{F2} (MPa)	134.994	152.004
Allowable bending stress on teeth of the wheel, σ_{FP2} (MPa)	508.354	505.959
Bending stress on teeth of the pinion, σ_{F3} (MPa)	143.11	170.473
Allowable bending stress on teeth of the pinion, σ_{FP3} (MPa)	573.289	563.03
Bending stress on teeth of the wheel, σ_{F4} (MPa)	155.327	147.2
Allowable bending stress on teeth of the wheel, σ_{FP4} (MPa)	501.088	509.001
Minimum fatigue life safety factor on the input shaft, F_{sfs1}	3.479	2.825
Minimum fatigue life safety factor on the intermediary shaft, F_{sfs2}	3.682	3.609
Minimum fatigue life safety factor on the output shaft, F_{sfs3}	2.921	3.492
Maximum equivalent von Misses stress for input shaft, σ_{echs1} (MPa)	42.684	58.534
Maximum equivalent von Misses stress for intermediary shaft, σ_{echs2} (MPa)	46.385	46.777
Maximum equivalent von Misses stress for output shaft, σ_{echs3} (MPa)	61.723	50.791
Estimated speed-reducer operating temperature, t_o (°C)	66.3	72.1
Speed reducer overall dimensions, $w \times L \times h$	418 × 360 × 225	377 × 326 × 201
Speed reducer mass (kg)	41.355	33.933

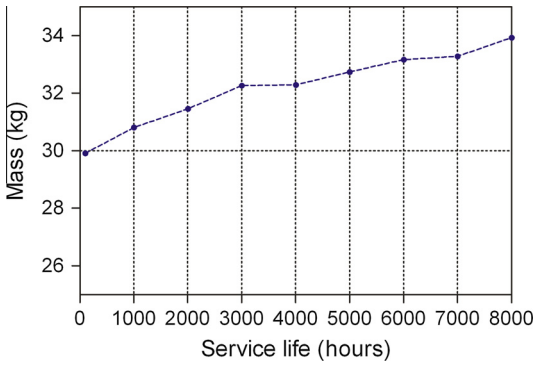


Fig. 5. The values of the constraints of the optimization problem at the optimum design.

be based on an ISO 53 basic rack profile ($\alpha_n = 20^\circ$, $h_{an} = 1$, $c_{sa} = 0.4$) with the pinions and wheels made of case hardened alloy steel 17CrNiMo6 and 17Cr3, respectively. Running the proposed Genetic Algorithm described earlier in Section 3 led to a speed reducer with a 3.15 and 2.24 standardized transmission ratio (on 1st and 2nd stage, respectively), a center distance of 90 mm on both stages, **weighing 33.933 kg**. The values of all considered genes, after optimization, are given in Table 2.

In Table 3 the main geometrical elements of the gearings of the speed reducer, for traditional [17,19,20] and optimal design solutions are shown side-by-side. In Tables 4 and 5 the shafts and also some important characteristics of the speed reducer for both design solutions are compared. As it can be observed from Table 5, the optimal design solution offers a mass reduction of 17.94% as opposite to the traditional design method [17,19,20]. If a large series of production is considered, the advantages are obviously, and the manufacturing costs are significantly diminished. For example, at ≈5 speed reducers produced, 1 is for free taking into account the material. In Fig. 5 the values of the constraints for the optimum design solution are represented. The black dots highlight the values of the constraints and with '×' symbol are indicated the 7 constraints whose boundaries are closest to that optimum.

In contrast to the successful determination of this global optimum (Fig. 5) consider the interesting study of a unique insight into the trade-off between the *mass* (i.e. the objective function in our case) and the *service life*. The result of such a study is plotted in

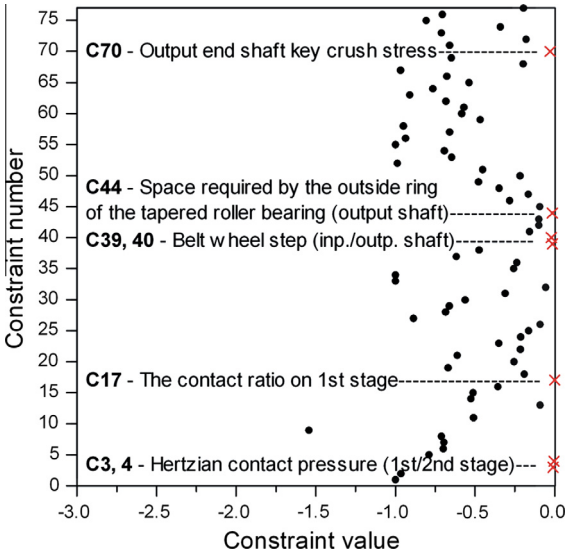


Fig. 6. Minimum mass speed reducers designed for various service life spans.

Fig. 6. From this interesting case study we can drawn the following conclusions (as are shown on the plot from Fig. 5). Halving the required service life would only save us just over 1 kg. If we were aiming for, say, 2000 h, we would need to sacrifice 75% of this service life for a ~2.5 kg (roughly 7%) weight saving. At the same time, even if weight was our sole concern, and we drop under 2000 h the mass saving is to less as compared to huge 75% sacrifice of service life.

6. Conclusions

Speed reducers are the most common ways of transmitting power used in mechanical engineering from different fields and in various working conditions. Nevertheless, traditional design (i.e. a commonly trial and cut error procedure) of such mechanical transmission is not an easy task.

In this paper we have shown how a GA, can be used to solve a complex structural design problem of a two-stage coaxial helical speed reducer. The **complete optimization of the speed reducer**

described here represents the state of the art of a study carried out along a few phases. During this preliminary steps were performed optimizations for the **gearings** [14] and the **shafts** [15]. The purpose of these phases was to determine the dimensional tendencies of the speed reducer's components. After finalizing these steps, the authors conclude that **is mandatory** to consider **the gearings**, **the shafts sub-assembly** and **the housing** with **all the interconnections between them resulting a whole system**. Only in this case we can speak about **a complete optimization of the reducer** with relevant results.

In this study the objective of the optimal design was the minimization of the complete speed reducer mass (including here even the housing which is designed automatically once the virtual design of each stage is generated). The design variables considered in the optimization are of mixed nature i.e., integer (e.g. the gears number of teeth), discrete (e.g. normal tooth addendum coefficients) and real (e.g. gears width), in a total of 17. The objective function was subjected to a highly non-linear set of 76 constraints. The results obtained by using GA show significant improvement over the results obtained by traditional design (a 17.94% mass reduction). Also an interesting study about the trade-off between the mass and the service life was presented. From this study we can conclude that the required service life has to be sacrificed with 75% for a ≈2.5 kg saving (roughly ≈7%).

The case study presented in this paper represents an optimal design of a new complex layout of a two-stage speed reducer necessary to *build a generic transmission system design tool based on the evolutionary optimization concepts* [1]. The proposed GA could be easily modified to suit multi-objective design optimization. Also additional stages and other objective functions (the manufacturing cost is merely a possible example) could be considered. This optimization example illustrates the effectiveness of the proposed approach and also serves as further evidence of the power and versatility of GAs in designing mechanical power transmissions.

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